Varad Rane | *Platform / Cloud Infrastructure Engineer*

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Summary

Platform engineer, 3 years building and operating backend infrastructure in a regulated production environment. Owned alert management across 100+ production servers and message queues, an internal API gateway centralizing auth and rate limiting, and the shared data-access library 50+ engineers depended on at one of India's largest brokerages. Built a fully serverless AWS document pipeline - Step Functions, Lambda, Bedrock, Aurora/pgvector — defined end to end in modular Terraform across separate dev and prod environments, deployed by GitHub Actions via OIDC federation with no static credentials. Also took an LLM assistant to production on Azure. Comfortable debugging production failures down to the socket layer.

Experience

Software Engineer II Jun 2022 - Aug 2024
Kotak Securities Mumbai, India

- Architected alert management across 100+ production servers and message queues with configurable health checks and thresholds, cutting quarterly downtime from 6 hours to 5 and automating post-incident root-cause reporting.
- Diagnosed and resolved a production IIS socket-exhaustion failure that had been degrading service reliability; eliminated two network hops per request in the process.
- Designed an internal API gateway centralizing authentication, CORS and rate limiting, with SQL-driven routing configuration that enabled parallel live-environment testing against production data with no separate staging deployment.
- Built token-based authentication from scratch for trading APIs serving 2M users at ~260 req/s peak, 30% above the previous 200 req/s, establishing the security and audit-compliance baseline for all later internal applications.
- Standardized authentication across backend services by integrating third-party 2FA into an internal IAM layer; partnered with external penetration testers to remediate 12 findings and authored the resulting security compliance documentation.
- Built a cross-compatible .NET Standard 2.1 shared library for data access and logging, adopted by 50+ engineers across 4 teams.
- Migrated a legacy CRM (70 views) from .NET Framework to .NET 7 with zero production downtime, resolving breaking changes and realigning auth flows to updated security standards.
- Optimized high-frequency production SQL via execution-plan analysis and indexing, cutting p95 latency on critical retrieval paths from 237 ms to 145 ms.

Software Developer Feb 2026 - Present
Field Worker Princeton, NJ

- Took a local RAG prototype to production on Azure (AI Foundry GPT-4.1, PostgreSQL/pgvector): an agentic case-summary assistant inside the Node.js API with 10 tool-calling functions for per-patient summaries, risks, urgent flags and caseload-wide reviews, plus its React chat UI.
- Built a nightly QA compliance sweep pairing deterministic SQL checks for unit and form gaps with an LLM verdict on whether each task narrative justifies what was billed; the model never does arithmetic, and results persist for a compliance dashboard.
- Architected a local-first inference pipeline (Ollama, ChromaDB, PostgreSQL vector storage) engineered to run inside a hard 8GB VRAM budget so regulated data never left the environment; adopted for 97% of new client entries in production.
- Designed a reproducible benchmarking harness measuring latency, throughput and retrieval quality across chunking, retrieval and quantization strategies, turning a configuration choice into a measured decision; raised retrieval accuracy from 72% to 93%.
- Built a document-extraction pipeline turning multi-column, cross-page PDFs into schema-validated JSON, automating ~95% of client intake.

Teaching Assistant Sep 2025 - Dec 2025
New Jersey Institute of Technology Newark, NJ

- Built an automated grading system executing untrusted student Python and C++ submissions against test suites in isolated containers, with LLM-generated structured feedback; cut turnaround for 50 students from 5 days to 6 hours.

Projects

Cloud Document Extraction Pipeline
https://github.com/varadrane7/cloud-extraction-pipeline
AWS Lambda, Step Functions, Bedrock, Aurora + pgvector, S3, Terraform, GitHub Actions

- Fully serverless, event-driven pipeline: S3 ingest triggers a Step Functions state machine with Distributed Map for page-level parallelism across 8 Lambdas (ingest validation, page rendering, vision transcription, schema mapping, validation/scoring, embedding, cost tracking, API serving), with retry/catch and an SQS dead-letter queue on failure.

- **Infrastructure as code**: entire stack in modular Terraform with separate dev and prod environments, IAM roles, state machine, VPC, Aurora Serverless v2, Lambda packaging, API Gateway.
- **CI/CD**: GitHub Actions deploying to AWS via OIDC federation with no static credentials; automated lint, test and `terraform validate` gated on every pull request.
- **Observability & cost control:** real-time cost-per-document tracking through CloudWatch Embedded Metric Format across Bedrock tokens, Lambda compute and S3 storage.
- **Data layer:** Aurora PostgreSQL + pgvector with an HNSW index for cosine-similarity semantic search; Pydantic v2 entity modeling with field-level confidence scoring.

NJ Transit Bus ETA Service

Python, GTFS-Realtime (protobuf), Docker, Home Assistant

- Containerized service that polls NJ Transit's GTFS-Realtime feed every 30 s, joins it with the static GTFS schedule, and serves per-stop ETAs over a read-only REST API with health and debug endpoints.
- Three-tier fallback per bus: realtime trip predictions, then GPS estimates (vehicle position projected onto the route polyline at timetable pace), then the timetable; explicit stale / no_upcoming / error states instead of silent wrong answers.
- Hardened like production: auto-refreshing auth tokens, cached static feed, read-only root filesystem, all Linux capabilities dropped, 256 MB memory cap; drives Home Assistant "leave now" alerts for a 4-person apartment.

PrivGuard Sentinel

https://blog.varadrane.com/posts/privguard-sentinel

Python, GitHub Actions, LLM APIs

- CI-integrated LLM pull-request reviewer running as a GitHub Action, posting inline comments from structured JSON with configurable severity thresholds.

Education

New Jersey Institute of Technology — Newark, NJ
MS, Computer Science - GPA 3.90 / 4.00

University of Mumbai — Mumbai, India
BE, Electronics Engineering - GPA 9.32 / 10

Skills

Cloud & IaC: AWS (Lambda, Step Functions, Bedrock, Aurora Serverless v2, S3, API Gateway, CloudWatch, IAM, VPC, SQS), Terraform (modular, multi-environment), infrastructure as code, serverless architecture, event-driven architecture, OIDC federation, Azure (AI Foundry, Container Registry, Web Apps), Microsoft Entra ID / MSAL
CI/CD & Containers: Docker, GitHub Actions, CI/CD pipelines, keyless deploy via OIDC, automated lint/test/terraform-validate gating, build automation, container isolation, Linux, Bash
Reliability & Observability: monitoring, alerting, health checks, incident response, post-incident root-cause analysis, CloudWatch, Embedded Metric Format, cost observability, dead-letter queues, retry/backoff, structured logging, performance profiling, p95/p99 latency analysis
Security & Access: OAuth 2.0, JWT, 2FA, IAM, rate limiting, API gateway policy, penetration-test remediation, security compliance documentation
Platform & Backend: REST API design, API gateways, microservices, distributed system design, ASP.NET Core, .NET Core/Standard, Node.js, node-schedule, GTFS-Realtime, protobuf
Data: PostgreSQL, SQL Server, Aurora, pgvector, query optimization, execution-plan analysis, indexing, schema migration
Languages & Scripting: Python, Go, Bash, C#, SQL/T-SQL, TypeScript

Certificates and Publications

AWS Building Production-Ready AI Agents with Amazon Bedrock AgentCore (2026)